GENOMELENS

Economic analysis of a genomic report

Cost–utility model of acting on the findings in one whole-genome input, valued at $100,000 per QALY from a healthcare sector perspective.

Source input	**public benchmark genome NA12878**	Input type	**WGS**
Generated	**24 October 2026**	Time horizon	**each decision model states its own**
Discounting	**3.0%, costs and QALYs**		

Proprietary.
Demonstration report for a public research genome. GenomeLens, its economic models and its methods are proprietary. This report shows results, not the method: it is not open-source software, not a description sufficient to reproduce the analysis, and not a license to use either.

Public benchmark genome.
A public benchmark genome. It is a real person's genome; the findings are shown to demonstrate the analysis and are not a clinical report.

Research and educational use — not medical or financial advice.
This is a decision-analytic model, not a clinical or economic evaluation submitted to any authority. Its parameters are population averages applied to one genome: it values the expected consequences of acting on a pattern of findings, and does not predict what will happen to any individual. Monetary figures include health valued at a willingness-to-pay threshold and are not cash returned to anyone. Clinically relevant findings require confirmation by an appropriate professional — a physician, clinical pharmacist or board-certified genetic counselor — before anything follows from them.

Economic impact overview

Modeled economic impact

MODELED NET MONETARY BENEFIT (SCENARIO)

$39

PRIMARY MODELED NET MONETARY BENEFIT UNDER THE STANDARDIZED AGE-45 SCENARIO

ASSUMPTION-DRIVEN SCENARIO

CYP2C19 Intermediate Metabolizer

Modeled decision pathway: genotype-guided escalation

$2,104 once the decision arises (per person for whom it arises) CONTEXT CONDITIONAL · EVIDENCE-QUALIFIED ESTIMATE; the expected value above weights it by the chance that the decision arises.

Lifetime value = $4.40 (the next 10 years) ASSUMPTION-DRIVEN SCENARIO + $35 (after the first 10 years, to age 110) ASSUMPTION-DRIVEN SCENARIO: the two windows of one decision, each counted once.

+0.000616 QALYs	+$22	Assumed age 45	$100,000 / QALY
HEALTH GAIN	INCREMENTAL COST	SCENARIO	THRESHOLD

MODEL EVIDENCE
Supported: every model input sourced at the assumed age

DONOR APPLICABILITY
Not established with available context: the donor's age is not observed

Net monetary benefit: the QALYs gained valued at $100,000 each, less the change in healthcare cost; a health-economic value, not cash saved. perspective: US; horizon: lifetime (the next 10 years and the years after); costs in 2019 US dollars.

Cost-effectiveness: $35,958 per QALY gained, below the $100,000 threshold.

Estimand: returning an already generated result.

Other modeled values (shown separately, never added)

ONCE THE DECISION ARISES

CONTEXT CONDITIONAL · ASSUMPTION-DRIVEN SCENARIO

$2,522

CYP2C19 Intermediate Metabolizer → ticagrelor-aspirin for loss-of-function carriers. Per person for whom it arises; not an expected value today

OTHER PRIMARY MODELED VALUE

ASSUMPTION-DRIVEN SCENARIO

$2,101

lifestyle intervention + medical therapy when a top-decile PRS lifts an intermediate-clinical-risk (FINDRISC ten-to-twenty-percent band) adult to a ten-year type 2 diabetes risk of twenty percent or more (Martikainen strategy). Its own decision or window, valued once

2 further modeled values are in the table that follows, each with its class.

SUPPORTED CALCULATION AT THE ASSUMED AGE

Modeled net monetary benefit at assumed age 45: **$4.40** ASSUMPTION-DRIVEN SCENARIO — the supported calculation at the assumed age: only decisions whose every model input is sourced. A stricter test, not a total of the values above, and it does not replace them.

Research and educational use only; not medical, clinical or financial advice. Standardized economic scenarios: age 45, using genomic findings identified from publicly available research datasets. Genomic calls in this proof of concept have not been independently confirmed by a clinical laboratory and are not diagnostic results. Its assumptions are listed on the economic methodology sheet.

All economically relevant findings

Every economic result of this genome, ranked within its class. Values in different classes are never added.

FINDING	DECISION PATHWAY	MODELED NMB	EVIDENCE CLASSIFICATION	KEY LIMITATION
lifestyle intervention + medical therapy when a top-decile PRS lifts an intermediate-clinical-risk (FINDRISC ten-to-twenty-percent band) adult to a ten-year type 2 diabetes risk of twenty percent or more (Martikainen strategy)		**$2,101**	PRIMARY MODELED · ASSUMPTION-DRIVEN SCENARIO	Counted: the selected model's expected value
refer an overweight (BMI 25-29.9) top-decile BMI-PRS adult to a counselor-led intensive lifestyle program (USPSTF refers only BMI>=30)		**$189**	PRIMARY MODELED · ASSUMPTION-DRIVEN SCENARIO	Counted: the selected model's expected value
CYP2C19 Intermediate Metabolizer	genotype-guided escalation	**$39**	PRIMARY MODELED · ASSUMPTION-DRIVEN SCENARIO	Counted: the decision's lifetime expected value (its two windows added once)
CYP2C19 Intermediate Metabolizer	ticagrelor-aspirin for loss-of-function carriers	**$2,522** per person for whom it arises	STANDARDIZED MODELED · CONTEXT CONDITIONAL · ASSUMPTION-DRIVEN SCENARIO	Once the decision arises; its expected value today, weighted by the chance it arises, is the primary row
CYP2C19 Intermediate Metabolizer	genotype-guided escalation	**$2,104** per person for whom it arises	STANDARDIZED MODELED · CONTEXT CONDITIONAL · EVIDENCE-QUALIFIED ESTIMATE	Once the decision arises; its expected value today, weighted by the chance it arises, is the primary row
CYP2C19 finding	Pharmacogenomic-guided antidepressant selection (Ghanbarian 2023, 20-year model)	**per patient tested**	TESTING-STRATEGY / POPULATION-LEVEL VALUE	A population testing-strategy average per patient tested, not this genotype's value

All economically relevant findings (continued)

FINDING	DECISION PATHWAY	MODELED NMB	EVIDENCE CLASSIFICATION	KEY LIMITATION
CYP2C19 finding	Pharmacogenomic-guided antidepressant selection (PREPARE psychiatry estimate)	**per patient tested**	TESTING-STRATEGY / POPULATION-LEVEL VALUE	A population testing-strategy average per patient tested, not this genotype's value
CYP2C9 finding	Genotype-guided warfarin dosing (CYP2C9, VKORC1)	**per patient tested**	TESTING-STRATEGY / POPULATION-LEVEL VALUE	A population testing-strategy average per patient tested, not this genotype's value
Hypothetical, based on assumptions: any adult, before results	Population screening for Lynch syndrome, hereditary breast and ovarian cancer and familial hypercholesterolemia	**per patient tested**	TESTING-STRATEGY / POPULATION-LEVEL VALUE	A population testing-strategy average per patient tested, not this genotype's value
ANKK1		**valued at zero**	TRUE ZERO	ISPG 2019: common psychiatric risk variants are individually of no clinical significance and should not be tested for …
CYP2C19 (Pharmacogenomic-guided antidepressant selection (Ghanbarian 2023, 20-year model))		**valued at zero**	TRUE ZERO	No change in care for this genotype: CPIC 2023 SSRI guideline …
CYP2C19 (Pharmacogenomic-guided antidepressant selection (PREPARE psychiatry estimate))		**valued at zero**	TRUE ZERO	No change in care for this genotype: CPIC 2023 SSRI guideline …
CYP2C19 Intermediate Metabolizer (IM) (clopidogrel)		**valued at zero**	TRUE ZERO	No change in care for this genotype: CPIC 2023 SSRI guideline …
CYP3A5 Non-expressor (PM) (tacrolimus)		**valued at zero**	TRUE ZERO	The CPIC tacrolimus guideline starts a non-expresser (*3/*3) at the standard recommended dose, so this genotype does …
DRD2		**valued at zero**	TRUE ZERO	ISPG 2019: common psychiatric risk variants are individually of no clinical significance and should not be tested for …
HLA-DQA1		**valued at zero**	TRUE ZERO	USPSTF 2017: insufficient evidence to screen asymptomatic people for celiac disease (I statement)

All economically relevant findings (continued)

FINDING	DECISION PATHWAY	MODELED NMB	EVIDENCE CLASSIFICATION	KEY LIMITATION
HLA-DQB1		**valued at zero**	TRUE ZERO	USPSTF 2017: insufficient evidence to screen asymptomatic people for celiac disease (I statement)
UGT1A1 Intermediate Metabolizer (IM) (irinotecan)		**valued at zero**	TRUE ZERO	This phenotype does not change the prescribing decision
ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding)		**valued at zero**	TRUE ZERO	Not a pathogenic variant. ATP7B c.2495A>G p.Lys832Arg (rs1061472), the only variant behind this row, is classified ...
vitamin D (at age 45)		**valued at zero**	TRUE ZERO	Outside the trial's population at this age: VITAL enrolled men 50 and older and women 55 and older ...
CYP2C9 Intermediate Metabolizer (IM) (warfarin)		**not quantified**	UNQUANTIFIED	Actionable finding, person-level value not quantified: the CPIC warfarin guideline (2017, PMID 28198005) gives this ...
SLCO1B1 Intermediate Function (IM-like) (simvastatin)		**not quantified**	UNQUANTIFIED	The CPIC statin guideline (Cooper-DeHoff 2022, PMID 35152405) adjusts the statin and dose for decreased or poor ...
CYP2D6 Intermediate Metabolizer (IM) (codeine / antidepressants)		**not assessable**	TECHNICALLY UNRESOLVED	CYP2D6 copy number not assessed: a SNV/indel callset cannot see the *5 deletion or gene duplications ...

All economically relevant findings (continued)

The evidence hierarchy

- **Evidence-supported**: every model input is sourced and the value does not rest on an age that is stated rather than observed (badge EVIDENCE-QUALIFIED)
- **Primary modeled**: the selected economic model for a decision; it may depend on the stated assumptions shown beside it **in this genome**
- **Once the decision arises**: the value per person if the decision arises; not an expected value today **in this genome**
- **Alternative modeled**: a competing model or sensitivity scenario for the same decision; never added to the primary result
- **Conditional**: valid only if a stated clinical or treatment condition holds; never added
- **Family / cascade-testing**: accrues to relatives or another beneficiary; never in the individual's personal total
- **Population / testing strategy**: a value per person screened or tested in a population, not this donor's value; never added **in this genome**
- **Hypothetical / projected**: an exploratory scenario on stated assumptions, not an observed or established benefit; in no total **in this genome**
- **Negative**: modeled, and the costs exceed the health gain valued at $100,000 per QALY; shown, never hidden
- **True zero**: the evidence supports no incremental benefit or change in care within the defined decision **in this genome**
- **Unquantified**: potentially relevant, but the evidence is insufficient for a monetary value; not $0 **in this genome**
- **Technically unresolved**: the genomic evidence cannot reliably support the interpretation the decision needs; not $0 **in this genome**
- **Not applicable**: the model or intervention does not apply to the assessed context (sex, age or setting); not $0

Unknown is not $0. Unquantified is not $0. Technically unresolved is not $0. No supported estimate is not a measured zero. Negative is not zero. An alternative is never added.

Economic methodology: how the primary modeled value is calculated

FINDING
CYP2C19 Intermediate Metabolizer

DECISION / INTERVENTION
genotype-guided escalation; compared with universal clopidogrel

MODEL EVIDENCE
Supported: every model input sourced at the assumed age

DONOR APPLICABILITY
Not established with available context: the donor's age is not observed; the figure is the standardized age-45 scenario, a standardized calculation, not a clinical prediction for this donor

CHANGE IN HEALTH OUTCOME
+0.000616 QALYs

CHANGE IN COST
+$22 (costs in 2019 US dollars; positive is more spending)

ECONOMIC CALCULATION
0.00062 QALYs times $100,000 per QALY, minus the change in cost of $22

RESULT

$39 ASSUMPTION-DRIVEN SCENARIO

Primary modeled net monetary benefit

Source and scope

population: patients with acute coronary syndrome undergoing PCI (Limdi 2020); per carrier: the per-patient value over the 32% who carry a loss-of-function allele; valued per: carrier for whom the decision arises; setting: US; horizon: lifetime (the next 10 years and the years after); sources: PMID 32042096, PMID 27751900, PMID 41848042. Net monetary benefit is the QALYs gained valued at $100,000 each, less the change in healthcare cost, discounted at 3% a year unless the row names its source's own rate.

Key assumptions

- at the stated age of 45, assumed for this run, not observed
- age stated, not observed
- age stated, not observed; the clopidogrel share held at its latest printed value for life

Where multiple economic models are available, GenomeLens evaluates evidence relevance, population applicability, intervention comparability, parameter provenance, and uncertainty. Alternative supported models may be retained as sensitivity scenarios.

Findings valued by their decision models

Net monetary benefit (NMB) is the QALYs gained valued at $100,000 each, less the change in healthcare cost, in 2019 US dollars. It is a health-economic value, not cash saved: a positive NMB can come with higher spending. Each clinical decision is valued once; the categories are never added to each other. Each figure's horizon is its own model's (the carrier models run over a lifetime; no pooled disease model ran for this genome).

Standardized economic scenarios: age 45, using genomic findings identified from publicly available research datasets. Genomic calls in this proof of concept have not been independently confirmed by a clinical laboratory and are not diagnostic results. Age 45 is stated for this run, not observed (one demonstration age for every sample; the PGP public profiles give only an age bracket, and the GIAB and 1000 Genomes samples publish no age). A value whose model depends on age is modeled at that age; a value its source gives for one age only is held at that age, and its row says so. Model evidence is stated apart from whether its context matches the donor, and the qualification stays assumption-driven; an age-dependent value would be evidence-qualified only at an observed or self-reported age.

Sex: known female.

Supported calculation at assumed age 45

$4.40 ASSUMPTION-DRIVEN SCENARIO · AGE 45 ASSUMED the supported calculation: every model input sourced, for a person of the assumed age; the draws put it between -$0.96 and $9.82 (5th to 95th percentile of the draws). Whether it applies to this donor is not established: the donor's age is not observed.

- **CYP2C19-guided antiplatelet therapy after a coronary stent**: $4.40 ASSUMPTION-DRIVEN SCENARIO

  expected ΔQALY +6.867e-05; expected ΔCost $2.47 (2019 USD; positive is more spending); ΔLY not reported by the source; 5th to 95th percentile of the draws -$0.96 to $9.82 (parameter uncertainty only).

  If the decision arises (per person for whom it arises): $2,104 CONTEXT CONDITIONAL · EVIDENCE-QUALIFIED ESTIMATE; ΔQALY +0.03285; ΔCost $1,181; its own inputs: derived from sourced inputs. Chance it arises: 0.002413, the next 10 years; the expected value above is that chance times this value, discounted to when it arises.

  Model evidence: supported: every model input sourced at the assumed age; **donor context:** not established with available context: the donor's age is not observed.

  Age restriction: the chance it arises is taken from the source's age bands.

  Holds at the stated age of 45, assumed for this run, not observed.

  Limited by: age stated, not observed.

  comparator: universal clopidogrel; window in which the decision may arise: the next 10 years; sources: PMID 32042096, PMID 27751900, PMID 41848042.

A. Donor-specific economic value

Not established with available context — the donor's age is not observed, and 6 priced decision(s) depend on it; no total is stated for this donor, which is not $0.

- none

B. Other modeled expected value, with further limitations

Further-limited subtotal $2,356 ASSUMPTION-DRIVEN SCENARIO — limited by inputs besides the age; never added to the supported calculation above.

- **lifestyle intervention + medical therapy when a top-decile PRS lifts an intermediate-clinical-risk (FINDRISC ten-to-twenty-percent band) adult to a ten-year type 2 diabetes risk of twenty percent or more (Martikainen strategy)**: $2,101

  ASSUMPTION-DRIVEN SCENARIO

  ΔQALY not reported separately by the source; ΔCost inside the source's estimate, not reported separately; ΔLY not reported by the source.

  Model evidence: limited: not evidence-supported even at an observed age; **donor context:** not established with available context: the donor's age is not observed.

  Holds at the stated age of 45, assumed for this run, not observed.

  Limited by: the score changes this decision; the score changes this decision; age stated, not observed.

  sources: PMID 36186460.

Findings valued by their decision models (continued)

B. Other modeled expected value, with further limitations (continued)

- **refer an overweight (BMI 25-29.9) top-decile BMI-PRS adult to a counselor-led intensive lifestyle program (USPSTF refers only BMI>=30)**: $189 ASSUMPTION-DRIVEN SCENARIO

  ΔQALY not reported separately by the source; ΔCost inside the source's estimate, not reported separately; ΔLY not reported by the source.

  Model evidence: limited: not evidence-supported even at an observed age; **donor context:** not established with available context: the donor's age is not observed.

  Holds at the stated age of 45, assumed for this run, not observed.

  Limited by: the score changes this decision; the score changes this decision; age stated, not observed.

  sources: PMID 31773873.

- **CYP2C19-guided antiplatelet therapy after a coronary stent (after the first 10 years)**: $35 ASSUMPTION-DRIVEN SCENARIO

  expected ΔQALY +0.0005477; expected ΔCost $20 (2019 USD; positive is more spending); ΔLY not reported by the source; 5th to 95th percentile of the draws -$9.21 to $78 (parameter uncertainty only).

  If the decision arises (per person for whom it arises): $2,104 CONTEXT CONDITIONAL · EVIDENCE-QUALIFIED ESTIMATE; ΔQALY +0.03285; ΔCost $1,181; its own inputs: derived from sourced inputs. Chance it arises: 0.03752, after the first 10 years, to age 110; the expected value above is that chance times this value, discounted to when it arises.

  Model evidence: limited: not evidence-supported even at an observed age; **donor context:** not established with available context: the donor's age is not observed.

  Holds at the stated age of 45, assumed for this run, not observed.

  Limited by: age stated, not observed; the clopidogrel share held at its latest printed value for life.

  comparator: universal clopidogrel; window in which the decision may arise: after the first 10 years, to age 110; sources: PMID 32042096, PMID 27751900, PMID 41848042.

- **CYP2C19-guided antiplatelet therapy after a minor stroke or TIA (after the first 10 years)**: $26 ASSUMPTION-DRIVEN SCENARIO

  expected ΔQALY +0.0001842; expected ΔCost -$7.28 (2019 USD; positive is more spending); ΔLY not reported by the source; 5th to 95th percentile of the draws $15 to $47 (parameter uncertainty only).

  If the decision arises (per person for whom it arises): $2,522 CONTEXT CONDITIONAL · ASSUMPTION-DRIVEN SCENARIO; ΔQALY +0.01982; ΔCost -$540; its own inputs: modeled with stated assumptions. Chance it arises: 0.03525, after the first 10 years, to age 110; the expected value above is that chance times this value, discounted to when it arises.

  Model evidence: limited: not evidence-supported even at an observed age; **donor context:** not established with available context: the donor's age is not observed.

  Holds at the stated age of 45, assumed for this run, not observed.

  Limited by: age stated, not observed; QALYs lost per recurrent stroke at 65; QALY per extra minor bleed; the DAPT share held at its latest printed value for life.

  comparator: clopidogrel-aspirin; window in which the decision may arise: after the first 10 years, to age 110; sources: PMID 34708996, PMID 38181311, PMID 38533952.

Findings valued by their decision models (continued)

B. Other modeled expected value, with further limitations (continued)

- **CYP2C19-guided antiplatelet therapy after a minor stroke or TIA**: $5.71 ASSUMPTION-DRIVEN SCENARIO

  expected ΔQALY +4.771e-05; expected ΔCost -$0.94 (2019 USD; positive is more spending); ΔLY not reported by the source; 5th to 95th percentile of the draws $3.43 to $11 (parameter uncertainty only).

  If the decision arises (per person for whom it arises): $2,522 CONTEXT CONDITIONAL · ASSUMPTION-DRIVEN SCENARIO; ΔQALY +0.01982; ΔCost -$540; its own inputs: modeled with stated assumptions. Chance it arises: 0.002065, the next 10 years; the expected value above is that chance times this value, discounted to when it arises.

  Model evidence: limited: not evidence-supported even at an observed age; **donor context:** not established with available context: the donor's age is not observed.

  Holds at the stated age of 45, assumed for this run, not observed.

  Limited by: age stated, not observed; QALYs lost per recurrent stroke at 65; QALY per extra minor bleed.

  comparator: clopidogrel-aspirin; window in which the decision may arise: the next 10 years; sources: PMID 34708996, PMID 38181311, PMID 38533952.

C. Values that hold only in a stated clinical context

- none for this genome

D. Alternatives and scenarios, shown separately

- none for this genome

E. Actionable pathways not yet quantified

- CYP2C9 Intermediate Metabolizer (IM) (warfarin) — actionable economically unquantified: the CPIC warfarin guideline (2017, PMID 28198005) gives this CYP2C9/VKORC1 genotype a different starting dose from the standard dose, but the economic evaluation (Meckley 2010) is an average per patient starting warfarin, every genotype's dose changed, with no result by genotype, so no value is estimated for this genotype (shown only as a testing-strategy value).
- SLCO1B1 Intermediate Function (IM-like) (simvastatin) — actionable economically unquantified: the CPIC statin guideline (Cooper-DeHoff 2022, PMID 35152405) adjusts the statin and dose for decreased or poor SLCO1B1 function. The only randomized trial of SLCO1B1-guided prescribing (I-PICC, PMID 33270123) tested LDL-C non-inferiority across all randomized patients, most of them non-carriers; it is not a measured null for a carrier, and no cost-utility estimate of the myopathy it averts exists.
- CYP2D6 Intermediate Metabolizer (IM) (codeine / antidepressants) — technically unresolved: CYP2D6 copy number not assessed: a SNV/indel callset cannot see the *5 deletion or gene duplications (the CYP2D7 pseudogene region), so this call is technically unresolved; only a poor-metabolizer call from two no-function alleles, which neither changes, is kept.

F. Valued at zero: care does not change (a result, not a gap)

- **ANKK1** — valued at zero: ISPG 2019: common psychiatric risk variants are individually of no clinical significance and should not be tested for (https://ispg.net/genetic-testing-statement/)
- **CYP2C19 (Pharmacogenomic-guided antidepressant selection (Ghanbarian 2023, 20-year model))** — valued at zero: no change in care for this genotype: CPIC 2023 SSRI guideline (Bousman 2023, PMID 37032427; vendored PA166127638, PA166127639) starts a CYP2C19 normal, rapid or intermediate metabolizer on citalopram, escitalopram or sertraline at the recommended starting dose, so this genotype does not change the modeled decision; the modeled values (Ghanbarian 2023; PREPARE) are averages per patient tested with a CYP2D6 + CYP2C19 panel, with no result by genotype, and are shown only as testing-strategy values

Findings valued by their decision models (continued)

F. Valued at zero: care does not change (a result, not a gap) (continued)

- **CYP2C19 (Pharmacogenomic-guided antidepressant selection (PREPARE psychiatry estimate))** — valued at zero: no change in care for this genotype: CPIC 2023 SSRI guideline (Bousman 2023, PMID 37032427; vendored PA166127638, PA166127639) starts a CYP2C19 normal, rapid or intermediate metabolizer on citalopram, escitalopram or sertraline at the recommended starting dose, so this genotype does not change the modeled decision; the modeled values (Ghanbarian 2023; PREPARE) are averages per patient tested with a CYP2D6 + CYP2C19 panel, with no result by genotype, and are shown only as testing-strategy values
- **CYP2C19 Intermediate Metabolizer (IM) (clopidogrel)** — valued at zero: no change in care for this genotype: CPIC 2023 SSRI guideline (Bousman 2023, PMID 37032427; vendored PA166127638, PA166127639) starts a CYP2C19 normal, rapid or intermediate metabolizer on citalopram, escitalopram or sertraline at the recommended starting dose, so this genotype does not change the modeled decision; the modeled values (Ghanbarian 2023; PREPARE) are averages per patient tested with a CYP2D6 + CYP2C19 panel, with no result by genotype, and are shown only as testing-strategy values
- **CYP3A5 Non-expressor (PM) (tacrolimus)** — valued at zero: the CPIC tacrolimus guideline starts a non-expresser (*3/*3) at the standard recommended dose, so this genotype does not change the dose
- **DRD2** — valued at zero: ISPG 2019: common psychiatric risk variants are individually of no clinical significance and should not be tested for (https://ispg.net/genetic-testing-statement/)
- **HLA-DQA1** — valued at zero: USPSTF 2017: insufficient evidence to screen asymptomatic people for celiac disease (I statement); ACG 2023 recommends against mass screening, HLA typing being used to rule celiac out (PMID 28350936; PMID 36602836)
- **HLA-DQB1** — valued at zero: USPSTF 2017: insufficient evidence to screen asymptomatic people for celiac disease (I statement); ACG 2023 recommends against mass screening, HLA typing being used to rule celiac out (PMID 28350936; PMID 36602836)
- **UGT1A1 Intermediate Metabolizer (IM) (irinotecan)** — valued at zero: this phenotype does not change the prescribing decision
- **ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding)** — valued at zero: not a pathogenic variant. ATP7B c.2495A>G p.Lys832Arg (rs1061472), the only variant behind this row, is classified Benign in ClinVar (Variation 35707) and is a common polymorphism, not a Wilson disease allele (ClinVar Variation 35707).
- **vitamin D (at age 45)** — valued at zero: outside the trial's population at this age: VITAL enrolled men 50 and older and women 55 and older (LeBoff 2022, PMID 35939577); Endocrine Society 2024 (Demay, PMID 38828931): for adults younger than 50 it suggests against empiric vitamin D beyond the dietary reference intake and against routine 25(OH)D testing

Findings-first economics

No pooled result: no finding enters the pooled model, so there is no combined estimate to state; each finding is valued by its own decision model, by category, on the next page.

This report starts with what was found in the genome. The economics explains the consequence of acting on each finding; the pooled reference case shows what remains after overlapping signals, imperfect adherence and the real cost of acting are all accounted for.

HEALTH GAIN

Not available

no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost; the findings are valued by their own decision models instead, by category, in the section 'Findings valued by their decision models'

HEALTHCARE COST CHANGE

Not available

no pooled result

MODELED NET BENEFIT

Not available

no pooled result

DECISION UNCERTAINTY

Not available

no pooled result

0

findings carry a standardized economic estimate

31 of the 31 findings shown carry no standardized estimate, and none is assigned one the model cannot support. Separately, 6 clinical decision(s) are valued by their own decision models (0 evidence-qualified, 6 modeled with stated assumptions), by category in the section 'Findings valued by their decision models'; a finding with such a value has its figure there, not here.

DRD2/ANKK1 DRD2 / ANKK1 Taq1A · Vitamin D Synthesis Efficiency · HLA-DQ2/DQ8 carrier · Wilson Disease · Celiac-permissive HLA haplotype… · CYP2C19 Intermediate Metabolizer · UGT1A1 Intermediate Metabolizer · Elevated breast-cancer polygenic… · Elevated obesity-tendency… · Elevated type-2-diabetes… · CYP2D6 Intermediate Metabolizer · CYP3A5 Non-expressor · CYP2C9 Intermediate Metabolizer · SLCO1B1 Intermediate Function · GABRA2 · CHRNA5 · CRHR1 — Stress-axis variant · Reproductive genetics · Oxidative Stress Defense · PAH carrier · PLG pathogenic · TPP1 pathogenic · and 9 more

Medication-genotype findings

Where a genotype could change a prescribing decision

These are the clearest cases of a genomic result bearing on a future medication choice. The clinical decision leads; the economics explains why preserving the genotype for that decision has value.

This run produced no medication-genotype finding with a priced economic pathway; a finding whose decision the evidence values at zero is listed with the true zeros.

Decision support, not a directive. Nothing here is an instruction to start, stop or change a medication. Relevance depends on whether the drug is actually prescribed, and the prescribing decision belongs with a qualified clinician who can see the whole clinical picture.

Risk and prevention findings, in detail

No risk or prevention finding of this genome has a standardized estimate, so there is no card to show. The findings are reported without one below, each with its reason on the glance pages; a lower evidence grade changes how a finding should be read, not whether it is shown.

Reported without a standardized value (31): DRD2/ANKK1 DRD2 / ANKK1 Taq1A..., Vitamin D Synthesis Efficiency..., HLA-DQ2/DQ8 carrier — celiac rule-out..., ATP7B rs1061472 (benign polymorphism)..., Celiac-permissive HLA haplotype..., CYP2C19 Intermediate Metabolizer (IM)..., and 25 more. Each carries its reason on the glance pages (no change in care, not assessable, unquantified, or reproductive information); no value is invented.

How the findings combine

Taken together, what do these findings mean?

There is no pooled reference case for this genome: no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost. Findings bearing on one condition are combined here before any value is credited; with nothing in the pool there is nothing to combine, and no figure stands in for the absent result. The findings are valued by their own decision models instead, by category, in the section 'Findings valued by their decision models'.

INCREMENTAL HEALTHCARE COST

Not available

no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost; the findings are valued by their own decision models instead, by category, in the section 'Findings valued by their decision models'

INCREMENTAL QALYS

Not available

no pooled result

REFERENCE-CASE NMB

Not available

no pooled result

CLASSIFICATION

Not available

no pooled result

How sure is this, and what rests on judgment?

Uncertainty has two layers: the spread of modeled outcomes, and the provenance of the inputs producing that spread. A robust decision does not make its inputs precise, so both are shown.

Net monetary benefit across simulations

No probabilistic analysis of the pooled model: no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost. No probability of cost-effectiveness, interval or one-way swing is stated for it, because there is no pooled strategy to sample. The decision models carry their own parameter uncertainty (2,000 draws from the ranges the sources print, rescaled to a per-carrier value where a source prints another unit); where a decision has a range, its 5th to 95th percentile is printed beside that decision's value.

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Base case: the value the published evidence supports for this genome, or “not estimable” with the reason. Scenario: what GenomeLens computes on its own assumptions, graded by the quality of its inputs, with each assumption numbered in the log below. Scenario figures are not from published data for this genome and are not the evidence headline; the reproductive costs belong to a prospective child and are in no personal total. A row valued by its own decision model repeats that model's value as a pointer; the model's basis (published, or hypothetical) is stated beside the headline. This genome's assessment is incomplete (a finding is actionable but unquantified, technically unresolved, or lacks the evidence to value): that is not a conclusion that the genome has no further economic value.

Table 1. Results by finding

FINDING	BASE CASE (PUBLISHED EVIDENCE)	SCENARIO (GENOMELENS ASSUMPTIONS)	WHY THE BASE CASE IS WHAT IT IS
This genome, to the person sequenced (each finding once)	**Not established** Incomplete assessment: not zero	**$19,207** Sum of the rows not marked 'not summed': model and scenario values, never this donor's total	**no evidence-supported value; $2,361 from decision models that are not evidence-qualified; $16,846 from 2 projected finding(s); figures marked 'not summed', reproductive values and payer values are excluded**
DRD2/ANKK1 DRD2 / ANKK1 Taq1A — DRD2/ANKK1 Taq1A — reward-pathway variant	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care (a guideline: care does not change)
Vitamin D Synthesis Efficiency — Nutrient-metabolism variant — deficiency prevention	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	Care does not change under a guideline (outside the trial's population at this age: VITAL enrolled men 50 and older and women 55 and older … Sources reviewed: PMID 17634462; PMID 32533132; PMID 35727271
HLA-DQ2/DQ8 carrier — celiac rule-out capability	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care (a guideline: care does not change)
ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	No earlier figure; the decision models' verdict: valued at zero: not a pathogenic variant
Celiac-permissive HLA haplotype (DQ2.5/DQ8) — serology decision	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care (a guideline: care does not change)
CYP2C19 Intermediate Metabolizer (IM) (clopidogrel)	**See its decision row** Decision model	**$4.40** Decision model (not summed)	Valued by the decision model 'CYP2C19-guided antiplatelet therapy after a coronary stent' (modeled with stated assumptions)

Hypothetical, based on assumptions (continued)

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Table 1. Results by finding (continued)

FINDING	BASE CASE (PUBLISHED EVIDENCE)	SCENARIO (GENOMELENS ASSUMPTIONS)	WHY THE BASE CASE IS WHAT IT IS
UGT1A1 Intermediate Metabolizer (IM) (irinotecan)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	No earlier figure; the decision models' verdict: valued at zero: this phenotype does not change the prescribing decision
Elevated breast-cancer polygenic risk	**Not estimable** Not estimable	**$12,443** Placeholder estimate (assumption)	No base-case figure. The one polygenic score with a per-person, horizon-matched, US-dollar cost-utility analysis is coronary artery disease ...
Elevated obesity-tendency polygenic risk	**Not estimable** Not estimable	**$4,403** Placeholder estimate (assumption)	No base-case figure. The one polygenic score with a per-person, horizon-matched, US-dollar cost-utility analysis is coronary artery disease ...
Elevated type-2-diabetes polygenic risk	**See its decision row** Decision model	**$2,101** Decision model (not summed)	Valued by the decision model 'lifestyle intervention + medical therapy when a top-decile PRS lifts an intermediate-clinical-risk (FINDRISC ten-to-twenty-percent band) ...
CYP2D6 Intermediate Metabolizer (IM) (codeine / antidepressants)	**Not estimable** Not estimable	**Not estimable** Conditional on a genotype call this input cannot establish; no figure is shown	No earlier figure; the decision models' verdict: not assessable from this input: CYP2D6 copy number not assessed: a SNV/indel callset cannot see the *5 deletion or gene ...
CYP3A5 Non-expressor (PM) (tacrolimus)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	No earlier figure; the decision models' verdict: valued at zero: the CPIC tacrolimus guideline starts a non-expresser (*3/*3) at the standard recommended dose, so this ...
CYP2C9 Intermediate Metabolizer (IM) (warfarin)	**Not estimable** Not estimable	**$1.05** Published population average, applied to this genotype (not summed)	No earlier figure; the decision models' verdict: actionable, with no economic estimate yet: actionable finding, person-level value not quantified: the CPIC warfarin ... Source: PMID 20014877
SLCO1B1 Intermediate Function (IM-like) (simvastatin)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	No earlier figure; the decision models' verdict: actionable, with no economic estimate yet: actionable finding with no economic anchor yet: the CPIC statin guideline ...
GABRA2 — Alcohol-metabolism susceptibility variant	**Not estimable** Not estimable	**No model yet** No model yet	An illustrative panel row: shown, not priced
CHRNA5 — Nicotine-dependence susceptibility variant	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care
CRHR1 — Stress-axis variant (CRHR1/FKBP5)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Table 1. Results by finding (continued)

FINDING	BASE CASE (PUBLISHED EVIDENCE)	SCENARIO (GENOMELENS ASSUMPTIONS)	WHY THE BASE CASE IS WHAT IT IS
Reproductive genetics — 2 carrier condition(s) where partner testing would be informative	**Separate arm** Separate arm	**See reproductive table** Reproductive table	Costed per risk in the reproductive rows (a prospective child's value, kept out of the personal total)
Oxidative Stress Defense — Reduced antioxidant enzyme efficiency	**Not estimable** Not estimable	**No model yet** No model yet	No economic model covers this finding, and no figure is assigned in its place.
PAH carrier — heterozygote health	**Not estimable** Not estimable	**No model yet** No model yet	Heterozygote health effects are real but not costed per gene
PLG pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
TPP1 pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
predicted-pathogenic FAM43A (3:194687573)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor
predicted-pathogenic ARHGEF19 (1:16206949)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor
predicted-pathogenic PAM (5:103003035)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor
predicted-pathogenic FGD5 (3:14821133)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor
predicted-pathogenic AK4 (1:65218865)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor
predicted-pathogenic SLC36A3 (5:151296240)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor
predicted-pathogenic NHP2 (5:178153531)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor
predicted-pathogenic FBXL6 (8:144356368)	**Not estimable** Not estimable	**No model yet** No model yet	Predicted pathogenic by a sequence model in a gene with no modeled condition anchor
39 further predicted-damaging variant(s), not listed individually	**Not estimable** Not estimable	**No model yet** No model yet	This genome carries 47 predicted-damaging variants. The economics examines the first 40 and names up to 8 unanchored ones

Table 2. Reproductive risks identified, with their cost (per pregnancy, for a prospective child; ancestry used: European)

REPRODUCTIVE RISK	CHANCE	COST OF THE CHILD'S CONDITION	EXPECTED COST	TEST THAT RESOLVES IT	SOURCE OF THE COST
HLA-DQA1 DQ2.5 tag Celiac Disease Susceptibility; dominant; carrier	**1 in 2** that a child inherits it, per pregnancy	**$3,000** curated outcome cost for a child with one copy	**Not stated** per pregnancy	**$185** a known-variant test of the pregnancy or the child	GenomeLens registered estimate

State: the figures on this page are model outputs, labeled UNRESOLVED (not established for this donor) unless a row carries another label. Standardized scenario at assumed age 45: the donor-specific economic value is not established with available context.

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Table 2. Reproductive risks identified, with their cost (per pregnancy, for a prospective child; ancestry used: European) (continued)

REPRODUCTIVE RISK	CHANCE	COST OF THE CHILD'S CONDITION	EXPECTED COST	TEST THAT RESOLVES IT	SOURCE OF THE COST
ATP7B K832R Wilson Disease (variable); recessive; common polymorphism, not a disease allele	**not stated** of an affected child, per pregnancy	**Not available** this allele is commonly reported as a polymorphism rather than a disease allele (classified as uncertain significance in the carrier catalog): it cannot complete an affected genotype, so neither a chance of an affected child nor a cost of the child's condition is stated; there is no risk for a partner's test to resolve, so none is priced	— per pregnancy	— No test priced: no risk to resolve	GenomeLens registered estimate

Methods

How these figures were produced, and how to read them

A decision-analytic model values the expected consequences of acting on each finding in this genome. The figures are model outputs under stated assumptions, kept apart from clinical outcomes.

ANALYTIC FRAME

Perspective	**healthcare sector**
Comparator	**each decision model states its own**
Outcomes	**cost, QALYs, NMB**
Time horizon	**each decision model states its own**
Discounting	**3%, costs and QALYs**
Willingness to pay	**$100,000/QALY**
Currency	**USD, United States**
Uncertainty	**decision models: 2,000-draw parameter uncertainty; pooled model: no PSA (no pooled model)**

HOW TO READ EACH FIGURE

- **Evidence-supported value:** every model input is sourced and the value does not rest on an age that is stated rather than observed. Where the age is stated, as for every public genome, an age-dependent value is a standardized scenario and is labeled so.
- **Decision-model rows:** computed from published inputs with at least one assumption (for a public genome, the stated age); shown, never presented as evidence-qualified.
- **Hypothetical, based on assumptions:** what a finding could be worth if its curated assumptions held; never added to any counted total.
- **Projected, based on assumptions:** the GenomeLens scenario beside a base case that cannot be computed soundly; not the evidence headline.
- **Reproductive costs:** per pregnancy, for a prospective child, never in a personal total.

Declared limitations

- Population-average parameters are applied to one genome. The model values expected consequences for a person with this pattern of findings, not a prediction for any individual.
- Estimates are illustrative and decision-analytic. This is not a formal economic evaluation or a submission to any authority.
- Where a condition-specific baseline risk is unavailable a generic registered assumption is used; those assumptions dominate the sensitivity analysis and are labeled as such.
- Costs are restated to 2019 US dollars from each source's price year.
- Findings without a modeled decision pathway are excluded from net monetary benefit rather than assigned an invented value.
- Clinically relevant findings require confirmation by an appropriate professional before anything follows from them.

Proprietary method. Demonstration report for a public research genome. GenomeLens, its economic models and its methods are proprietary. This report shows results, not the method: it is not open-source software, not a description sufficient to reproduce the analysis, and not a license to use either.